

# Securitize Vault V2 and RWASegWrap Audit Summary

Prepared by Cyfrin

Version 2.2


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# 1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

# 2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Securitize Vault V2 and RWASegWrap. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.2 dated August 4, 2025, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

# 3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

# 4 Copyright and License

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# 5 Audit Overview

Over the course of 10 days, the Cyfrin team conducted an audit on the Securitize Vault V2 and RWASegWrap code provided by Securitize. The findings consist of 1 Medium & 7 Low severity issues with the remainder being informational and gas optimizations.

**Summary**

| | |
|---|---|
| Project Name | Securitize Vault V2 and RWASegWrap |
| Repository | bc-securitize-vault-sc |
| Commit | 0c179554f39c... |
| Fix Commit | 5105f03e9550... |
| Repository 2 | bc-rwa-seg-wrap-sc |
| Commit | 2cb4c9d990f9... |
| Fix Commit | 6322e36c86c0... |
| Audit Timeline | Jul 21st - Aug 1st, 2025 |
| Methods | Manual Review |

**Issues Found**

| Severity | Found | Resolved | Acknowledged |
|---|---|---|---|
| Critical Risk | 0 | 0 | 0 |
| High Risk | 0 | 0 | 0 |
| Medium Risk | 1 | 1 | 0 |
| Low Risk | 7 | 7 | 0 |
| Informational | 5 | 4 | 1 |
| Gas Optimizations | 3 | 2 | 1 |
| Total | 16 | 14 | 2 |

# 6 Audit Scope

The review has been scoped to the following repositories and contracts:

1. `bc-rwa-seg-wrap-sc`

```
RWASegWrap.sol
SecuritizeRWASegWrap.sol
SegregatedVault.sol
VaultDeployer.sol
SegregatedVaultDeployer.sol
SecritizeVaultDeployer.sol
```

2. `bc-securitize-vault-sc`

```
SecuritizeVaultV2.sol
```

# 7 Risk Classification

| | **Impact: High** | **Impact: Medium** | **Impact: Low** |
|---|---|---|---|
| **Likelihood: High** | Critical | High | Medium |
| **Likelihood: Medium** | High | Medium | Low |
| **Likelihood: Low** | Medium | Low | Low |

# 8 Protocol Summary

The Securitize RWA Segregated Vault system is a Phase 2 evolution of the Phase 1 design that implements a hybrid ERC-20/ERC-1155-like architecture. The system creates individual ERC-4626 compliant vaults per investor while maintaining DeFi fungibility through a wrapper orchestration layer.

## 8.1 Key Features

- Segregated Collateralization: Each investor gets their own vault, simplifying tracking and liquidation
- Hybrid Token Model: ERC-20 tokens for investor compatibility and ERC-1155 family tracking for DeFi protocols
- Dynamic Vault Deployment: Vaults are auto-deployed on an investor's first deposit
- Dual Liquidation Modes: Direct DSToken transfers or redemption to stablecoins

## 8.2 Components

1. **RWASegWrap (Wrapper Layer)**: Orchestrates individual investor vaults and provides an ERC-1155 interface for DeFi.
2. **VaultDeployer (Factory Layer)**: Deploys individual vaults using the beacon proxy pattern.
3. **SegregatedVault/SecuritizeVaultV2 (Vault Layer)**: Individual ERC-4626 compliant vaults per investor.

## 8.3 Key Processes

Deposit:

1. Investor calls `deposit()` on the wrapper
2. On the first deposit, a new vault is deployed via `VaultDeployer`
3. Fees are calculated and collected via `FeeManager`
4. Net assets are transferred to the vault
5. The vault mints shares to the investor
6. The wrapper tracks the balance via its ERC-1155 interface

Liquidation:

1. A DeFi protocol calls `liquidate()` with a share amount and vault ID
2. The vault converts shares to assets using the NAV rate
3. The liquidator's shares are burned
4. Assets are transferred (direct mode) or converted via the redemption contract (redemption mode)

# 9 Attestation

Cyfrin attests that it performed a security review of the Securitize Vault V2 and RWASegWrap code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-08-04-cyfrin-securitize-vaultv2-rwasegwrap-v2.2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=71c42c370a0376c29745c7f4e5bdd968ee37f4e921f4f2cb6b2e597e7ef5b1c5
F=2025-08-04-cyfrin-securitize-vaultv2-rwasegwrap-v2.2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.